

Yash Kulkarni

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PROFILE SUMMARY

Pre-final year B.Tech student (Mechanical Engineering, CGPA 7.65) transitioning into full-stack development and applied AI/ML. Architected production-grade systems using Python, React.js, Node.js, FastAPI, and XGBoost — from AI-powered healthcare triage to blockchain-enabled agri-tech. National-level hackathon finalist (Silicon Plains Applied AI Summit 2026, MumbaiHacks 2025); 130+ DSA problems solved; seeking software engineering or AI/ML internship roles.

EDUCATION

Chh. Shahu College of Engineering
Bachelor of Technology in Mechanical Engineering; CGPA: 7.65

Aurangabad, Maharashtra
Aug 2023 – May 2027

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, Java
Frontend: React.js, HTML5, CSS3, Vite, Three.js
Backend: Node.js, Express.js, Flask, FastAPI, RESTful APIs
Databases: MongoDB, MongoDB Atlas, ChromaDB
AI/ML: XGBoost, EfficientNet, YOLOv5, OpenCV, LangChain, RAG Pipelines, Scikit-learn
Tools & Technologies: Git, GitHub, Vercel, Docker, Celery, Redis, ESP32, IoT Integration
Core CS: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems

PROJECTS

OralAI – Oral Cancer Triage System Python, Flask, EfficientNet, XGBoost, MongoDB, React.js
– Architected AI-powered oral cancer triage platform combining EfficientNet image classifier with XGBoost lifestyle risk engine; produces four-level triage decisions for rural Maharashtra screening camps.

– Built concurrent batch processing pipeline (Flask, Celery, Redis) reducing manual triage time by ~70%; designed multi-tenant MongoDB architecture projected at **\$40K+ annual revenue** and ~60% reduction in unnecessary specialist referrals.

AgriLink – AI-Powered Agri Value Chain Platform React.js, Node.js, FastAPI, MongoDB, XGBoost, Socket.io

– Built 3 XGBoost ML models for price prediction ($R^2=0.898$, MAE 215/q), demand forecasting (**76.1% accuracy**), and export premium estimation across 9 countries on AGMARKNET dataset (737K rows).

– Implemented SHA-256 blockchain hash chain for tamper-proof lot traceability with QR verification; built farmer-to-buyer marketplace with JWT auth and real-time Socket.io updates.

Intelligent Pesticide Sprinkling System Node.js, MongoDB, YOLOv5, OpenCV, ESP32

– Engineered full-stack IoT system for automated plant disease detection achieving **85%+ accuracy** using YOLOv5 and OpenCV; integrated ESP32-CAM for autonomous image capture and pesticide dispensing, cutting manual inspection by ~50%.

– Built scalable REST APIs (Node.js, Express.js) for event logging, authentication, and real-time analytics dashboard serving **100+ concurrent users**.

React Portfolio Website | [GitHub](#) React.js, Vite, Vercel

– Developed high-performance responsive portfolio achieving **95+ Lighthouse score**; reduced initial bundle size by 40% with lazy loading, code-splitting, and React Hooks — first load under 2 seconds.

ACHIEVEMENTS

Silicon Plains Applied AI Summit 2026 (Healthcare Track): Finalist – OralAI selected among top teams nationally for AI-powered oral cancer triage.

MumbaiHacks 2025: Finalist among 500+ teams; **Academic Excellence:** 1st and 2nd year ME Topper, 7.6+ CGPA; **CP:** 200+ DSA problems on LeetCode, CodeChef, GFG.

Hackathons: Semi-finalist, Agastya Anveshana Innovation Challenge; Participant, Smart India Hackathon (College Level).

CERTIFICATIONS

NPTEL: Programming in Java, Database Management Systems, Soft Skills Development

Forge: TATA Data Visualisation Virtual Internship, J.P. Morgan Software Engineering Virtual Internship